

Ctrl+Backspacing a word with non-BMP characters in an edit box deletes additional characters

MC-265929 Input

Status	Resolved / Fixed
Affected	1.20.2, 23w42a, 23w44a
Fixed in	1.20.3 Pre-Release 1
Reported	2023-10-19

STEPS TO REPRODUCE

1. Launch Minecraft 1.20.2.
2. From the main menu, open Multiplayer.
3. Select Add Server.
4. Paste `■ ■■ ■■■ ■■■■ ■■■■■ ■■■■■■ ■■■■■■■` into the Server Address box.
5. Move the cursor to the end of the pasted string.
6. Press Ctrl+Backspace repeatedly.
7. Observe that deleting a word also deletes the preceding word instead of deleting only one word at a time.

OBSERVED BEHAVIOUR

Actual behavior

When the string contains non-BMP characters such as the Shavian characters in the reproduction, ****Ctrl+Backspace deletes more text than the immediately preceding word****. The first press removes the word at the cursor and also removes part or all of the preceding word. Repeated presses therefore skip over multiple words instead of deleting exactly one word per press.

This occurs because the word boundary is calculated as a UTF-16 code-unit index, while the deletion offset is applied as a code-point offset. Each non-BMP character occupies two UTF-16 code units but is traversed as one code point, causing the deletion range to extend too far toward the beginning of the text.

EXPECTED BEHAVIOUR

When Ctrl+Backspace is used in an edit box containing non-BMP characters, it should delete only the single preceding word, including its associated spaces, and place the cursor at the resulting word boundary. Repeated presses should remove one preceding word per press without deleting any additional preceding words or characters, preserving the remaining text exactly.

ENVIRONMENT

Minecraft Java Edition Versions: 1.20.2, 23w42a, 23w44a (includes snapshots)

ORIGINAL DESCRIPTION

Using Ctrl+Backspace to delete a word in an edit box results in additional characters being deleted if that word contains non-BMP characters.

Reproduction

- Open the screen to add a server
- Paste the string **■ ■■ ■■■ ■■■■ ■■■■■ ■■■■■■ ■■■■■■■** (consisting of Shavian characters) into the 'Server Address' box
- Make sure the cursor is at the end of the string
- Press Ctrl+Backspace repeatedly

Intended behavior: only one word is deleted at a time

Actual behavior: the preceding word is deleted as well

Code analysis

(Based on the 1.20.2 code, using Yarn version 1.20.2+build.1. Decompiled with Fabric's fork of CFR, with some editing from me.)

TextFieldWidget#keyPressed calls TextFieldWidget#erase when it detects a Backspace or Delete press. erase, in turn, calls eraseWords or eraseCharacters depending on whether Ctrl is pressed.

eraseWords is implemented as follows:

```
public void eraseWords(int wordOffset) {
    if (this.text.isEmpty()) {
        return;
    }
    if (this.selectionEnd != this.selectionStart) {
        this.write("");
        return;
    }
    // The argument passed in is in code units, not code points!
    this.eraseCharacters(this.getWordSkipPosition(wordOffset) - this.selectionStart);
}

public void eraseCharacters(int characterOffset) {
    if (this.text.isEmpty()) {
        return;
    }
    if (this.selectionEnd != this.selectionStart) {
        this.write("");
        return;
    }
    int pos = this.getCursorPosWithOffset(characterOffset);
    int from = Math.min(pos, this.selectionStart);
    int to = Math.max(pos, this.selectionStart);
    if (from == to) {
        return;
    }
    String newText = new StringBuilder(this.text).delete(from, to).toString();
    if (!this.textPredicate.test(newText)) {
        return;
    }
    this.text = newText;
    this.setCursor(from, false);
}

private int getCursorPosWithOffset(int offset) {
    // Util.moveCursor traverses the string in code points, not code units!
    return Util.moveCursor(this.text, this.selectionStart, offset);
}

public int getWordSkipPosition(int wordOffset) {
    return this.getWordSkipPosition(wordOffset, this.getCursor());
}

private int getWordSkipPosition(int wordOffset, int cursorPosition) {
```

```
return this.getWordSkipPosition(wordOffset, cursorPosition, true);  
}  
priva
```

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-265929>